

May 11, 2026

Dear Hiring Committee:

I'm writing to express my interest in applying for open position at your organization. I believe my work experience along with my academic background have provided me with the credentials needed to thrive as your employee.

Over the last thirty years, I have worked on R&D in data analysis and management fields including astronomy, climate change simulation, combustion modeling, cosmology, energy efficiency, fusion science, genomics, geology, high energy physics, networking, nuclear science, power grid electricity, and behavioral economics. After a competitive review process, I became the first Senior Computing Engineer at Lawrence Berkeley National Laboratory (LBNL) in 2013. I applied the research results to scientific data management, computing and storage problems, by developing high-frequency streaming data analysis methods, dynamic resource management, I/O optimization solutions for exascale HPC applications, and statistical modeling and machine learning methods for autonomous scientific data infrastructure. I also have worked on networking problems for many years in network performance diagnostics and modeling, and more recently networking projects from ESnet such as SENSE-NRM and in-networking caching. I led the system designs, development of algorithms and models, implementation of the systems, management of teams, and software package releases under open source license. My experiences from the various science disciplines and collaboration coupled with my educational background in statistics, applied mathematics and computer science have enhanced my analytical skills over the years, and enabled me to provide different ways of thinking and the problem-solving techniques unique to each problem.

I have extensive grant proposal writing experience and led projects from the U.S. Department of Energy (DOE) and National Science Foundation (NSF) as a lead Principal Investigator (PI) or Co-PI. I was also involved in technical program committees, steering and advisory committees for conferences, journal editorial boards, and review panels in data, cloud computing, HPC, and networking areas, and I have been organizing workshops as one of the chairs, in conjunction with ACM HPDC and IEEE ICDCS. I have actively collaborated with scientific communities, including contributions to the Nobel Peace Prize 2007 to the Intergovernmental Panel on Climate Change (IPCC) with climate data management and the Nobel Prize in Physics 2017 to physicists from the Laser Interferometer Gravitational-Wave Observatory (LIGO) collaboration project with the software product managing scientific data storage. I was one of the main authors and editors of the standard specification from Open Grid Forum, and contributed to over 350 paper publications and technical reports, a number of open source software packages, and 3 patented technologies. Through my collaboration experience and involvement with various committees, I have demonstrated my leadership, communication, and project management abilities that make me an ideal candidate for the position.

I strongly believe that I have the skills and experiences necessary to make a significant contribution to your organization, and I would greatly appreciate the opportunity to meet with you to further discuss my qualifications in more detail. I can be reached via email at ASim@LBL.Gov or by phone at (925) 400-7460. I look forward to hearing from you soon.

Thank you for your time and consideration.

Sincerely,

Alex Sim  
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925-400-7460